The Costs and Benefits of Agglomeration

LSE GY457 - AT 2025
Topic 3
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Course overview

- Block 1
  - Intro to Urban and Regional Economics and course overview
  - Topic 1: Regional and urban concentration forces
  - Topic 2: The empirics of agglomeration
  - **Topic 3: The costs and benefits of agglomeration**
- Block 2
  - Topic 4&5: Monocentric city model
  - Topic 6: Firm location choice
- Block 3
  - Topic 7: Neighborhoods and segregation
  - Topic 8: The vertical dimension of cities
  - Topic 9: Gentrification and suburbanization
  - Topic 10: Measuring cities with new technologies

Recap from last time

- Empirics of agglomeration
  - How to measure spatial concentration
  - Which industries concentrate and why
  - How large are spatial spillovers
- Estimates of effects of agglomeration on productivity
  - Doubling density increases area productivity by ~4% on average
  - Doubling density increases individual productivity by ~2%
- Urbanization vs. localization effects
  - Localization economies impact distribution of productivity levels
  - Urbanization economies matter for growth

Today: The costs and benefits of agglomeration

- The economic effects of density
  - How to measure cities
  - Costs and benefits of density
- Equilibrium city size
  - Optimal city size
  - Inefficiencies and policies
- Compensating differentials
  - Rosen-Roback framework

Cities and the “fundamental trade-off of urban economics”

- Cities offer agglomeration benefits to workers, firms, and consumers
  - Among others: productivity, innovation, better access to goods and services, reductions in travel needs, broader sharing of (scarce) urban amenities, etc. (Duranton and Puga, 2020)
- … As well as costs
  - Crowding and congestion
  - Higher exposure to pollution and diseases (historically very relevant)

Cities and the "fundamental trade-off of urban economics"

- Benefits and costs of density not necessarily well weighted by policymakers and planners, partly because they are not shared equally among individuals
  - e.g., city residents vs. suburban residents/outsiders
- Partly because benefits and costs may operate at different spatial and temporal scales
  - e.g., congestion pricing policies
- What do we know about this trade-off?

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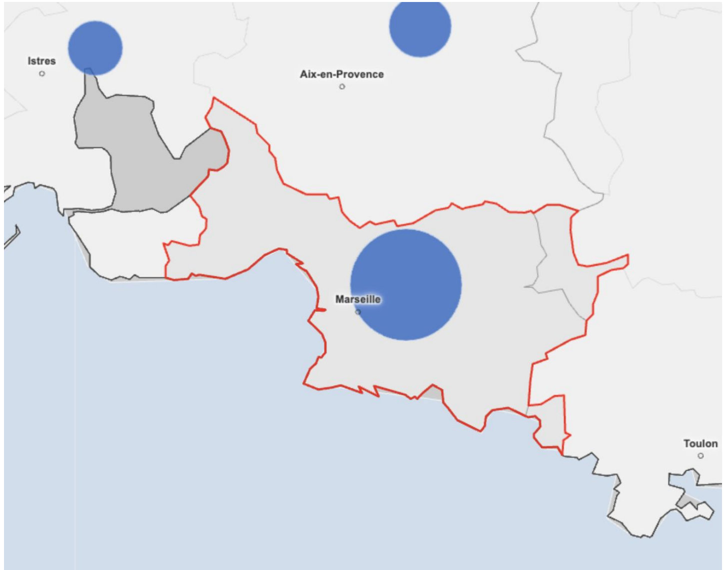
How do we measure density?

- How local is local?
  - Some characteristics could be measured in continuous space (e.g., employment density with individual-level data and flexible geography)
  - Others fundamentally depend on the unit (e.g., city population)
- Advancements in data processing and availability of granular data allow to work with continuous space
- To delineate urban space: think in terms of commuting flows (functional approach) or continuous built-up space (morphological approach)
  - No approach is conceptually better (depends on question at hand)

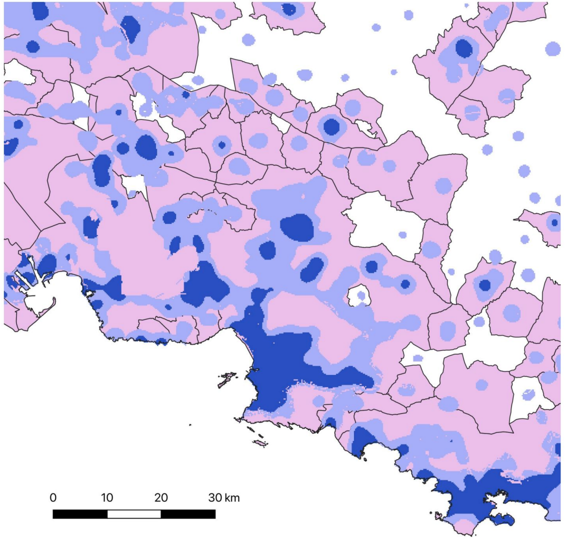
Example: Marseille

Marseille commuting zone



Source: INSEE (2020)

Marseille built-up area



Source: de Bellefon et al. (2021)

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The benefits from density: (i) Productivity benefits

- Central theme in urban economics
- Elasticity of productivity wrt density equal to 0.04 (meta-analysis of Ahlfeldt and Pietrostefani, 2019)
- Productivity and earnings higher in denser cities
  - Most estimates obtained by comparing productivity/earnings across locations with different densities
  - More recent studies use firm/worker-level data
- Elephants in the room: (location-specific) unobserved productivity advantages & workers' sorting
  - Omitted variable bias

(ii) Accessibility benefits

- Higher density implies shorter expected distances
  - But may encourage to more trips and hence slower travels
  - Duranton and Turner (2018): Elasticity of distance traveled to density equal to (only) -0.13; travel speed elasticity equal to -0.11, but also total time spent traveling declines (study of the US)
- Variety, prices, and quality of goods respond to density
  - Handbury and Weinstein (2015): Prices same (for same goods), but variety much higher; in larger areas, consumers buy higher quality goods (analysis of US markets using barcode data)
- Side note: Rich spatial and transportation data allows to compute various measures of accessibility

(iii) Among other benefits: Innovation, reduced pollution, amenities

- Many other benefits of agglomeration have been studied (can you guess others?)
- Higher concentration of innovative activities
- Lower greenhouse gas emissions and fewer particulates
  - Partly because of transport reductions, dwelling sizes
  - But: higher exposure to pollution (Carozzi and Roth, 2023)
- (Local) endogenous amenities partly determined by close proximity of high-skill residents (e.g., lower crime, more consumption opportunities)

The costs of density: (i) Land scarcity

- Higher density makes land scarce and housing more expensive
  - Higher land price is not an economic cost, but a transfer
  - But: it also reflects social costs
- More costly land gives incentives to build taller
  - Good: more floor space per unit of land
  - Bad: marginal cost of floor space increases with height
  - >> higher demand for land >> increase in floor space density >> higher prices for land and floor space >> lower consumption of floor space per person >> increase in human density per unit land

(ii) Congestion and travel costs

- As a city becomes denser and expands outwards, urban travel gets lower and congestion worsens
  - Elasticity of travel speed to city population equal to -0.04 in the US (Duranton and Puga, 2023); -0.05 in India (Akbar et al., 2019)
- City that expands makes travel longer and more costly
  - Trade-off between housing costs and transportation costs at the center of land-use modeling like the monocentric city model (more on this in the next class)
  - Locations further away have cheaper housing, but higher transportation costs (with the two forces offsetting each other)

(iii) Disamenities

- Crime often higher in denser cities
- Pollution and health-related issues
  - Historically, high density related to high premature deaths due to poor hygiene and faster spread of epidemics
  - Bairoch (1988): early in 19th century, rural youth expected to live 8-12 years longer than urban youth
  - In Europe and North America, urban life expectancy overtook rural life expectancy only in the 1930s

Netting out the costs and benefits of density in cities

- Inverse U-shape relationship between the net benefits of cities and the size of their population
  - As cities grow, the benefits of density seem to be higher than the associated costs of living in dense cities
  - Eventually, as cities become "too dense," the costs of density are higher than the benefits
- This relationship suggests that we could derive an *optimal city size* (at the peak)
  - At the same time, the peak appears to be quite flat, i.e. the costs of being moderately under or over-sized are small

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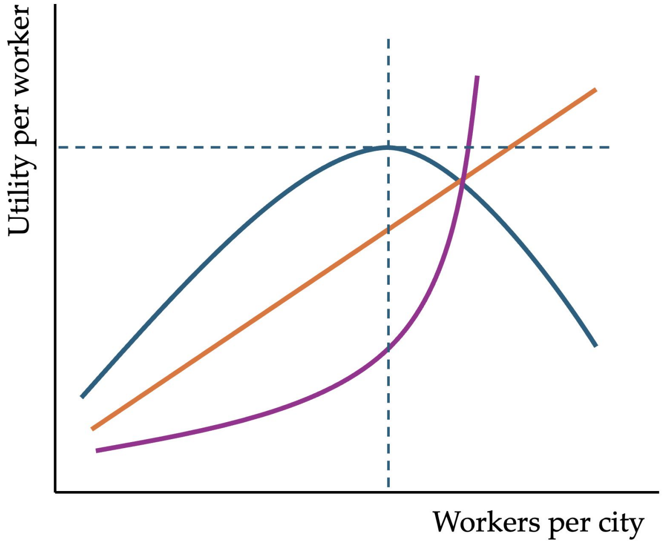
City size and migration

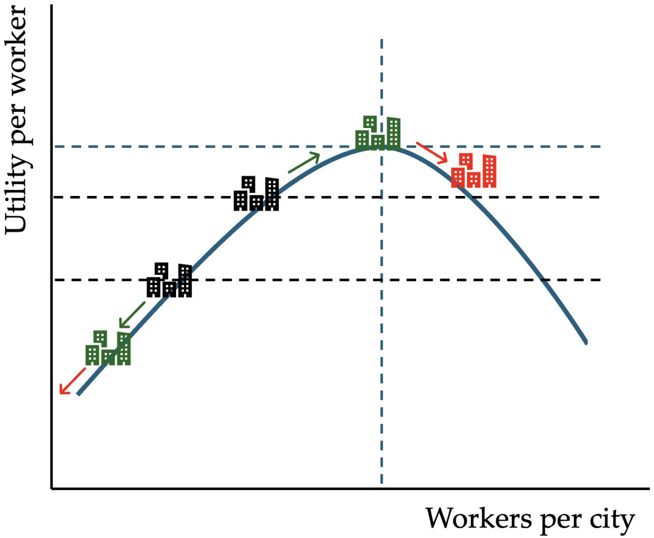
- Migration ensures that prices adjust to maintain spatial equilibrium
  - Eventually, city stops growing (as rents become too high)
  - Is the resulting equilibrium city size efficient?
- City size reflects trade-off between benefits and costs
  - Benefits: higher productivity and wages, consumption benefits (diversity, large-scale amenities,...)
  - Costs: high rents (scarcity of land), commuting costs, congestion, noise, pollution, crime,...
- As city grows, costs start outweighing the benefits
  - Does a city stop growing when too large, too small, or just right?

Equilibrium city size



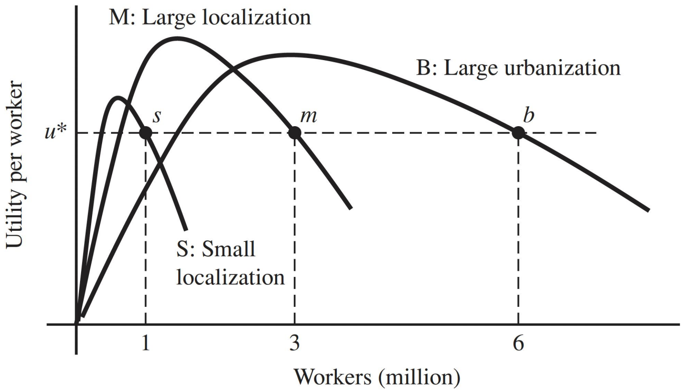
- Utility per worker increases in city size
  - Orange curve
- Disutility per worker increases in city size (at increasing rate)
  - Purple curve
- Net utility exhibits inverse U-shape
  - Blue (not same scale)

Towards optimal city size



- Net utility exhibits inverse U-shape
- Migration goes to larger cities with higher net-utility
- But: Does not stop at the optimum size
- Cities in equilibrium are likely too big!

System of cities



Source: O'Sullivan, Ch. 4

- Differences in city size
- Stable equilibrium (the utility per worker is the same in each city, i.e. u^*)
  - All cities are beyond optimal size > no incentive to migrate
- Can policies influence city size?

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Getting closer to optimal density?

- When making location decisions, firms and workers weight their personal (*private*) costs and benefits, not the *social* ones (generating externalities)
  a. Firms and workers consider agglomeration spillovers received from others, but not the ones they generate to others
  b. Residents do not internalize the extra costs they impose on others (added congestion), worsening as the city keeps expanding
  c. If land not owned by local residents, part of the benefits of density (agglomeration) are transferred away as rents
- Inefficient market outcomes
  - a & c >> suboptimally *low* density; b >> suboptimally *high* density

Policies and city size

- Land use regulations heavily affect housing supply
  - Why would we even need regulations? e.g. if localized costs vs. more diffuse benefits of new developments
  - More on this in winter term
- On the other hand, growth policies can positively influence city size
  - e.g., Improvements in urban infrastructure that reduce congestion costs (like new metro line); policies that attract sectors with high localization economies

Is there an optimal city size?

- Cities in equilibrium tend to be too big
- Denser cities seem to be welfare enhancing
- How can both be true?
- Costs and benefits of density are not necessarily shared by the same groups
  - Location decisions (migration) based on local costs
  - But: national/global benefits of density (more efficient public good provision, lower carbon emissions,...)
- Cities too big given local benefits, but possibly too small considering global benefits (Borck and Tabuchi, 2018)

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Compensating differentials: The Rosen-Roback model

- With free population mobility, in a spatial equilibrium utility is equalized across locations
  - Equilibrium >> Nobody has an incentive to move elsewhere
- What does this imply to rents and wages?
  - Rents and wages (endogenously) adjust to balance out (*compensate*) differences in desirability across locations

The Rosen-Roback framework

- Spatial equilibrium framework developed in Rosen (1979) and Roback (1982)
    - Workhorse tool to analyze local labor markets
- Consider a simplified version
    - Two cities $j = 1,2$
    - Two goods: traded composite good Q and (untradable) land H
    - Productivity (and utility) A_j differs exogenously across locations
    - Zero transport costs for trading good Q; price normalized to 1
    - Firms and workers are mobile and move between cities
    - Wages and rents adjust to clear markets
    - Spatial equilibrium implies that utilities and profits are constant

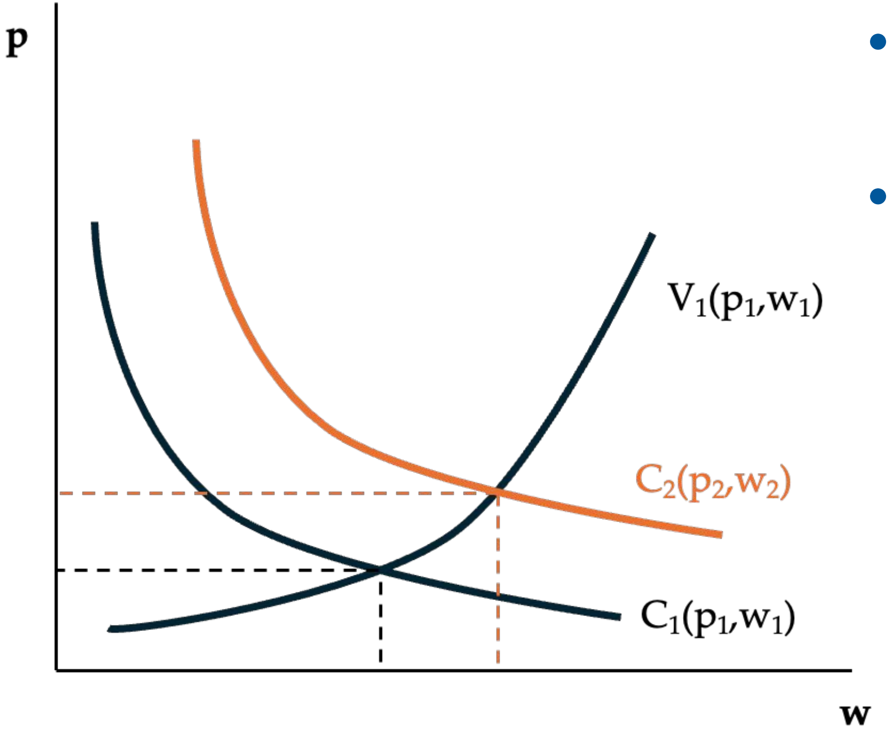
Rosen-Roback: Firms

- Firms produce good *Q* using labor *L* and land *H*
  - Output depends on (location) productivity: $Q_j = A_j(L_j, H_j)$
- Firms pay land rent *p* and wage *w*
  - Firm profits: $\pi_j = Q_j - p_j H_j - w_j L_j$
- Firms operate in perfectly competitive market
  - Competition implies zero profits
  - $1Q_j = A_j(L_j, H_j) = p_j H_j + w_j L_j$
  - Average cost of producing Q must be equal to its unit price (1)
  - >> Cost (and price) constant across cities: $C_1(p_1,w_1) = 1 = C_2(p_2, w_2)$
  - As *p* increases *w* must decrease and vice versa (what the firm spends in one factor must save on the other)

Rosen-Roback: Workers

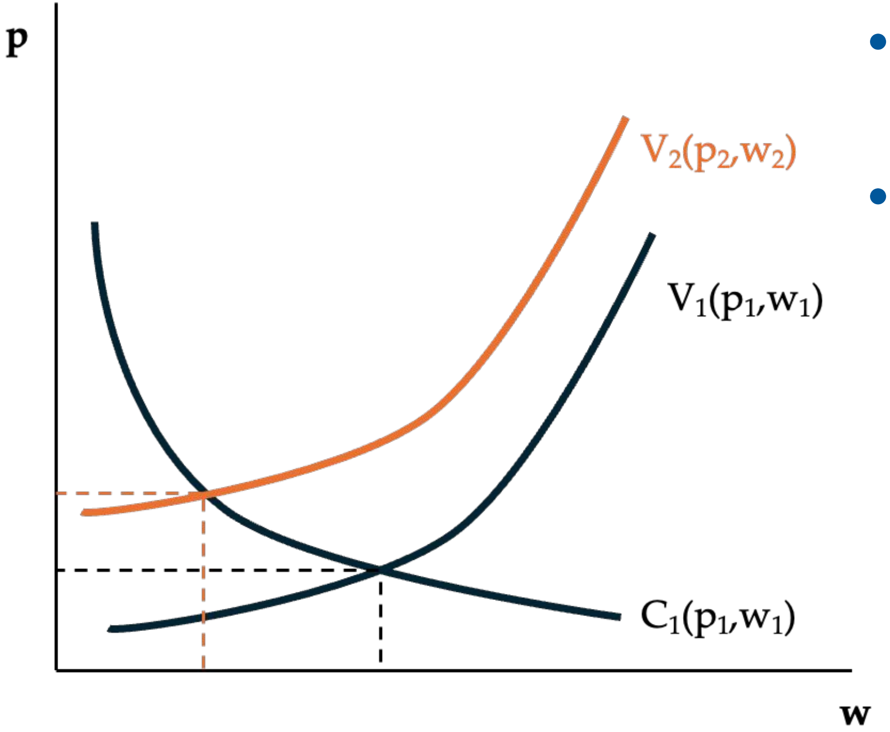
- Workers derive utility from consumption of the composite good q, land h, and (residential) amenity a
  - Lower case letters for workers: $U_j = U(a_j, q_j, h_j)$
- Workers spend income *w* to consume good *q* and land *h*
  - Budget constraint (BC): $w_j = 1q_j + p_j h_j$
- Maximization subject to BC gives indirect utility *V*
  - Exogenous utility fixed: $V_j = U(a_j, q^*_j, h^*_j) = U(a_j, q_j(p_j,w_j), h_j(p_j,w_j))$
  - Utility must be equalized across locations: $V_1(p_1,w_1) = V_2(p_2,w_2)$
  - As rent *p* increases, wage *w* must increase and vice versa

Rosen-Roback: Equilibrium (A↑)



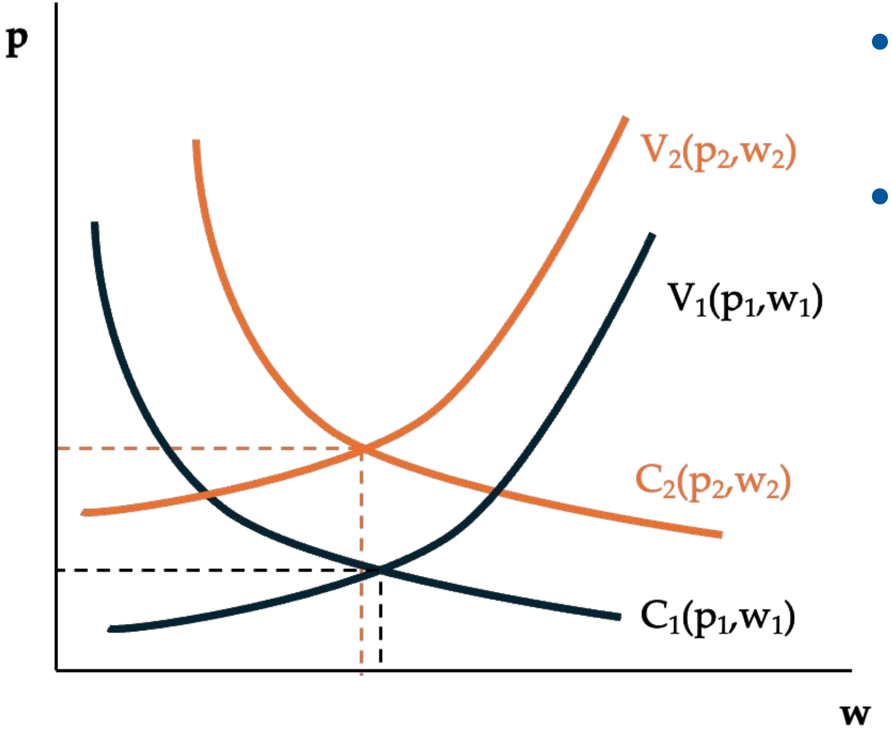
- Eq rents and wages ensure C and V are fixed (constant)
- With higher production amenity >> Higher rents and wages

Rosen-Roback: Equilibrium ($a\uparrow$)



- Eq rents and wages ensure C and V are fixed (constant)
- With higher residential amenity >> Higher rents and *lower* wages

Rosen-Roback: Equilibrium (A↑ & *a*↑)



- Eq rents and wages ensure C and V are fixed (constant)
- With higher production *and* residential amenity >> Higher rents and ambiguous effects on wages

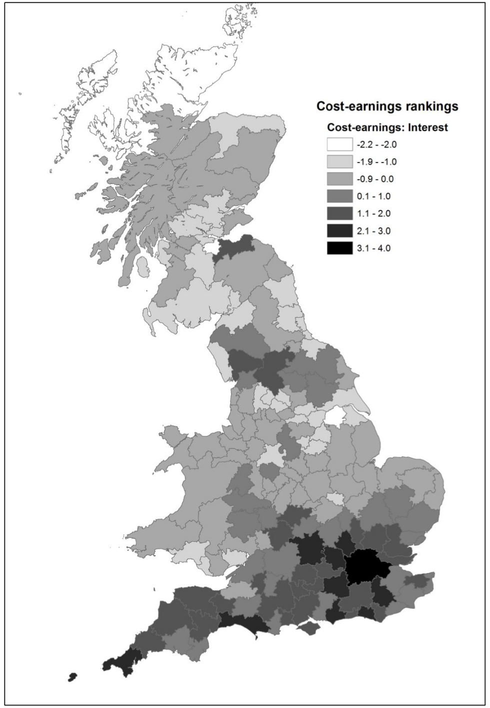
Rosen-Roback: Standard predictions

Residential amenities (e.g., scenic views)	Production amenities (e.g., subsidies from government)
Higher rents	Higher rents
Lower wages	Higher wages
• Residents willing to accept lower wages • Residents bid up rents • Firms must lower wages to keep profits constant	• Firms able to pay higher rents and wages • Firms bid up rents • Residents demand higher wages to keep utility constant

- Production (dis)amenities can be also consumption (dis)amenities
- Some amenities (local public goods) are funded by local taxes
- In the long run, residential amenities may lead to population concentration, induce agglomeration effects and increase rents (production amenity)
- Rent increase due to production amenities rests on the assumption of relevant crowding (relevant for various industries)

Evidence from Great Britain

- UK example (Gibbons et al., 2011)
- Top places: higher rents & lower wages
  - 1. London; 2. Brighton; 3. Guildford
- Figure shows spatial distribution of housing costs minus earning differentials across GB labor market areas
- Amenities (in places with lower real wages)
  - Employment access, jobs/household, woodland, museums, mean slope,...
- Disamenities (in places with higher real wages)
  - Rainfall, particulate matter, crime,...



Is density an amenity?

- Density increases wages and rents
  - >> Must be a production amenity
- Density increases rents more than wages
  - >> Must be a residential amenity
- But relationship between density and quality of life follows inverse u-shape
  - Density of zero and infinity seems undesirable
  - We don't know much about optimal density
  - Average city we observe somewhere below optimal density levels

Summary

- Cities and the “fundamental trade-off of urban economics”
    - Higher density associated with a range of costs and benefits
- Equilibrium city size
    - Cities tend to be too big given the local costs and benefits
    - …but potentially too small given the global externalities
- Compensating differentials
    - Productive cities have high wages and high rents
    - More amenable (livable) cities have higher rents relative to wages

Next week

- Introduction to monocentric city model
- Household bid-rent model
  - Simple Alonso (1964) model
  - Adding the construction sector: Mills (1967) and Muth (1969)
- Empirical patterns
  - (Urban) rent gradient

Thank you!

Readings and references

Required readings

- O'Sullivan, Ch. 4, City Size; Ch. 5, Urban Growth.
- Duranton, G., and Puga, D. (2020). The Economics of Urban Density. Journal of Economic Perspectives, 34(3), 3-26.

Other readings and references

- Ahlfeldt, G. M., and Pietrostefani, E. (2019). The Economic Effects of Density: A Synthesis. Journal of Urban Economics, 111, 93-107.
- Akbar, P., Couture, V., Duranton, G., and Storeygard, A. (2023). Mobility and congestion in urban India. American Economic Review, 113(4), 1083-1111.
- Albouy, D. (2008). Are big cities bad places to live? Estimating quality of life across metropolitan areas. National Bureau of Economic Research, No. w14472.
- Bairoch, P. (1988). *Cities and economic development: from the dawn of history to the present*. University of Chicago Press.
- Borck, R., and Tabuchi, T. (2018). Pollution and city size: can cities be too small? Journal of Economic Geography, 19(5), 995-1020.
- Carozzi, F., and Roth, S. (2023). Dirty density: Air quality and the density of American cities. Journal of Environmental Economics and Management, 118, 102767.

Readings and references

- Combes, P. P., Duranton, G., and Gobillon, L. (2019). The Costs of Agglomeration: House and Land Prices in French Cities. Review of Economic Studies, 86(4): 1556–1589.
- De Bellefon, M. P., Combes, P. P., Duranton, G., Gobillon, L., and Gorin, C. (2021). Delineating urban areas using building density. Journal of Urban Economics, 125, 103226.
- Duranton, G., and Puga, D. (2023). Urban growth and its aggregate implications. Econometrica, 91(6), 2219-2259.
- Gibbons, S., Overman, H. G., and Resende, G. (2011). Real earnings disparities in Britain. SERC Discussion paper 65.
- Handbury, J., and Weinstein, D. E. (2015). Goods prices and availability in cities. The Review of Economic Studies, 82(1), 258-296.
- Henderson, V. (1974). The Sizes and Types of Cities. The American Economic Review, Vol. 64, No. 4, pp. 640-656.
- Roback, J. (1982). Wages, Rents, and the Quality of Life. Journal of Political Economy, 90(6), 1257–1278.
- Rosen, S. (1974). Hedonic Prices and Implicit Markets: Product Differentiation in Pure Competition. Journal of Political Economy, 82(1), 34–55.

Appendix

Density elasticities in the literature

(Ahlfeldt and Pietrostefani, 2019)

	Elasticity of outcome		Proportion				Med.	Mean	Elasticity[g]	
ID	with respect to density	N	Poor[a]	Ac.[b]	Econ.[c]	With.[d]	Year[e]	SMS[f]	Mean	S.D.
1	Labour productivity	47	0.19	0.79	0.74	0.06	2007	3.02	0.04	0.04
1	Total factor productivity	15	0.13	0.87	0.80	0.20	2004	2.80	0.06	0.03
2	Patents p.c.	7	0.00	1.00	0.14	0.00	2006	2.86	0.21	0.11
3	Rent	13	0.00	0.69	0.62	0.62	2013	3.00	0.15	0.13
4	Commuting reduction	36	0.03	0.56	0.08	0.56	2005	2.17	0.06	0.12
4	Non-work trip reduction	7	0.00	0.71	0.00	0.86	2005	2.00	−0.20	0.44
5	Metro rail density	3	0.00	1.00	0.00	1.00	2010	3.33	0.01	0.02
5	Quality of life	8	0.38	0.88	1.00	0.13	2014	3.00	0.03	0.07
5	Variety (consumption amenities)	1	0.00	1.00	0.00	0.00	2015	4.00	0.19	–
5	Variety price reduction	2	0.00	0.00	1.00	1.00	2016	4.00	0.12	0.06
6	Public spending reduction	20	0.00	1.00	0.05	0.00	2007	2.00	0.17	0.25
7	90th-10th pct. wage gap reduction	1	0.00	1.00	0.00	0.00	2004	4.00	0.17	–
7	Black-white wage gap reduction	1	0.00	0.00	1.00	0.00	2013	2.00	0.00	–
7	Diss. index reduction	3	0.00	1.00	0.33	0.00	2009	3.33	0.66	0.94
7	Gini coef. reduction	1	0.00	1.00	0.00	0.00	2010	4.00	4.56	–
7	High-low skill wage gap reduction	3	0.00	0.67	1.00	0.00	2013	4.00	−0.13	0.07
8	Crime rate reduction	13	0.00	0.69	0.15	0.92	2014	2.54	0.24	0.47
9	foliage projection cover	1	0.00	1.00	0.00	1.00	2015	1.00	−0.06	–
10	Noise reduction	1	0.00	1.00	0.00	0.00	2012	1.00	0.04	–
10	Pollution reduction	18	0.44	0.33	0.33	0.39	2014	2.83	0.04	0.47
11	Energy reduction: Domestic & driving	21	0.10	0.90	0.38	0.24	2010	1.81	0.07	0.10
11	Energy reduction: Public transit	1	0.00	1.00	1.00	0.00	2010	1.00	−0.37	–
12	Speed	2	0.00	0.00	1.00	0.00	2016	4.00	−0.12	0.01
13	Car usage (incl. shared) reduction	22	0.00	0.95	0.00	0.95	2004	2.00	0.05	0.07
13	Non-car use	76	0.05	0.79	0.00	0.86	2006	2.03	0.16	0.24
14	Cancer & other disease reduction	5	0.00	1.00	0.00	0.60	2000	2.40	−0.33	0.20
14	KSI & casualty reduction	4	0.00	1.00	0.00	0.00	2003	2.00	0.01	0.61
14	Mental-health	1	0.00	1.00	0.00	1.00	2015	2.00	0.01	–
14	Mortality reduction	3	0.00	1.00	0.00	0.00	2010	2.00	−0.36	0.17
15	Reported health	3	0.00	1.00	0.00	0.00	2013	1.00	−0.27	0.11
15	Reported safety	1	0.00	1.00	0.00	1.00	2015	2.00	0.07	–
15	Reported social interaction	6	0.00	0.17	0.83	0.00	2007	3.50	−0.13	0.19
15	Reported wellbeing	1	0.00	1.00	1.00	0.00	2016	3.00	0.00	–
	Sum	347								